

Vector Spaces

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1 Vectors in R^n

- An ordered n -tuple :

a sequence of n real numbers $(x_1, x_2, \dots, x_n)$

- R^n -space :

the set of all ordered n -tuples

$n = 1$ R^1 -space = set of all real numbers

(R^1 -space can be represented geometrically by the x -axis)

$n = 2$ R^2 -space = set of all ordered pair of real numbers (x_1, x_2)

(R^2 -space can be represented geometrically by the xy -plane)

$n = 3$ R^3 -space = set of all ordered triple of real numbers (x_1, x_2, x_3)

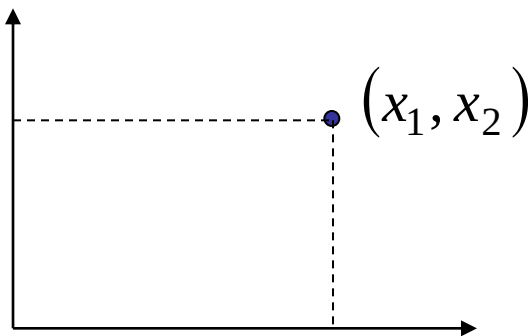
(R^3 -space can be represented geometrically by the xyz -space)

$n = 4$ R^4 -space = set of all ordered quadruple of real numbers (x_1, x_2, x_3, x_4)

- **Notes:**

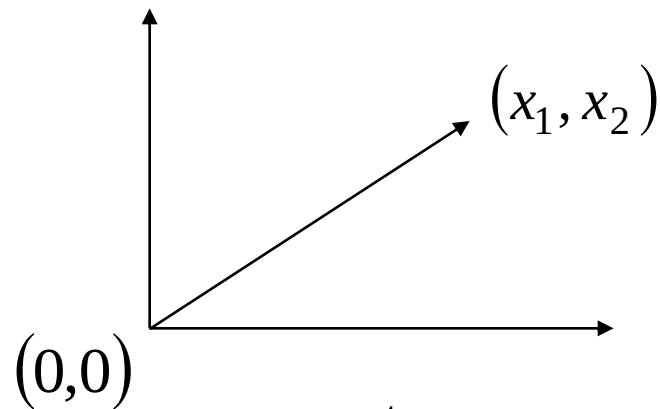
- (1) An n -tuple $(x_1, x_2, \dots, x_n)$ can be viewed as a point in R^n with the x_i 's as its **coordinates**
- (2) An n -tuple $(x_1, x_2, \dots, x_n)$ also can be viewed as a vector $\mathbf{x} = (x_1, x_2, \dots, x_n)$ in R^n with the x_i 's as its **components**

- **Ex:**



a point

or



a vector

$$\mathbf{u} = (u_1, u_2, \dots, u_n), \quad \mathbf{v} = (v_1, v_2, \dots, v_n) \quad (\text{two vectors in } R^n)$$

- Equality:

$$\mathbf{u} = \mathbf{v} \text{ if and only if } u_1 = v_1, u_2 = v_2, \dots, u_n = v_n$$

- Vector addition (the sum of $\mathbf{u}$ and $\mathbf{v}$):

$$\mathbf{u} + \mathbf{v} = (u_1 + v_1, u_2 + v_2, \dots, u_n + v_n)$$

- Scalar multiplication (the scalar multiple of $\mathbf{u}$ by c):

$$c\mathbf{u} = (cu_1, cu_2, \dots, cu_n)$$

- Notes:

The sum of two vectors and the scalar multiple of a vector in R^n are called the **standard operations in R^n**

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- Difference between **u** and **v**:

$$\mathbf{u} - \mathbf{v} \equiv \mathbf{u} + (-1)\mathbf{v} = (u_1 - v_1, u_2 - v_2, u_3 - v_3, \dots, u_n - v_n)$$

- Zero vector :

$$\mathbf{0} = (0, 0, \dots, 0)$$

■ Theorem 1: Properties of vector addition and scalar multiplication

Let $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ be vectors in R^n , and let c and d be scalars

- (1) $\mathbf{u} + \mathbf{v}$ is a vector in R^n (closure under vector addition)
- (2) $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$ (commutative property of vector addition)
- (3) $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$ (associative property of vector addition)
- (4) $\mathbf{u} + \mathbf{0} = \mathbf{u}$ (additive identity property)
- (5) $\mathbf{u} + (-\mathbf{u}) = \mathbf{0}$ (additive inverse property)
- (6) $c\mathbf{u}$ is a vector in R^n (closure under scalar multiplication)
- (7) $c(\mathbf{u} + \mathbf{v}) = c\mathbf{u} + c\mathbf{v}$ (distributive property of scalar multiplication over vector addition)
- (8) $(c + d)\mathbf{u} = c\mathbf{u} + d\mathbf{u}$ (distributive property of scalar multiplication over real-number addition)
- (9) $c(d\mathbf{u}) = (cd)\mathbf{u}$ (associative property of multiplication)
- (10) $1(\mathbf{u}) = \mathbf{u}$ (multiplicative identity property)

- Notes:

A vector $\mathbf{u} = (u_1, u_2, \dots, u_n)$ in R^n can be viewed as:

a $1 \times n$ row matrix (row vector): $\mathbf{u} = [u_1 \ u_2 \ \cdots \ u_n]$

or

a $n \times 1$ column matrix (column vector): $\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix}$

Vector addition

$$\begin{aligned}\mathbf{u} + \mathbf{v} &= (u_1, u_2, \dots, u_n) + (v_1, v_2, \dots, v_n) \\ &= (u_1 + v_1, u_2 + v_2, \dots, u_n + v_n)\end{aligned}$$

Scalar multiplication

$$\begin{aligned}c\mathbf{u} &= c(u_1, u_2, \dots, u_n) \\ &= (cu_1, cu_2, \dots, cu_n)\end{aligned}$$

Treated as $1 \times n$ row matrix

$$\begin{aligned}\mathbf{u} + \mathbf{v} &= [u_1 \ u_2 \ \dots \ u_n] + [v_1 \ v_2 \ \dots \ v_n] \\ &= [u_1 + v_1 \ u_2 + v_2 \ \dots \ u_n + v_n]\end{aligned}$$

$$\begin{aligned}c\mathbf{u} &= c[u_1 \ u_2 \ \dots \ u_n] \\ &= [cu_1 \ cu_2 \ \dots \ cu_n]\end{aligned}$$

Treated as $n \times 1$ column matrix

$$\mathbf{u} + \mathbf{v} = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix} + \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} = \begin{bmatrix} u_1 + v_1 \\ u_2 + v_2 \\ \vdots \\ u_n + v_n \end{bmatrix}$$

$$c\mathbf{u} = c \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix} = \begin{bmatrix} cu_1 \\ cu_2 \\ \vdots \\ cu_n \end{bmatrix}$$

2 Vector Spaces

- **Vector spaces :**

Let V be a set on which two operations (vector addition and scalar multiplication) are defined. **If the following ten axioms are satisfied** for every $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ in V and every scalar (real number) c and d , then V is called a **vector space**

Addition:

- (1) $\mathbf{u} + \mathbf{v}$ is in V
- (2) $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$
- (3) $\mathbf{u} + (\mathbf{v} + \mathbf{w}) = (\mathbf{u} + \mathbf{v}) + \mathbf{w}$
- (4) V has a zero vector $\mathbf{0}$ such that for every $\mathbf{u}$ in V , $\mathbf{u} + \mathbf{0} = \mathbf{u}$
- (5) For every $\mathbf{u}$ in V , there is a vector in V denoted by $-\mathbf{u}$ such that $\mathbf{u} + (-\mathbf{u}) = \mathbf{0}$

Scalar multiplication:

$$(6) \quad c\mathbf{u} \text{ is in } V$$

$$(7) \quad c(\mathbf{u} + \mathbf{v}) = c\mathbf{u} + c\mathbf{v}$$

$$(8) \quad (c + d)\mathbf{u} = c\mathbf{u} + d\mathbf{u}$$

$$(9) \quad c(d\mathbf{u}) = (cd)\mathbf{u}$$

$$(10) \quad 1(\mathbf{u}) = \mathbf{u}$$

※ Any set V that satisfies these ten properties (or axioms) is called a vector space, and the objects in the set are called vectors

※ Thus, we can conclude that R^n is of course a vector space

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- Four examples of vector spaces are introduced as follows. (It is straightforward to show that these vector spaces satisfy the above ten axioms)

(1) **n -tuple space: R^n**

$$(u_1, u_2, \dots, u_n) + (v_1, v_2, \dots, v_n) = (u_1 + v_1, u_2 + v_2, \dots, u_n + v_n) \text{ (standard vector addition)}$$

$$k(u_1, u_2, \dots, u_n) = (ku_1, ku_2, \dots, ku_n) \text{ (standard scalar multiplication for vectors)}$$

(2) **Matrix space :** $V = M_{m \times n}$

(the set of all $m \times n$ matrices with real-number entries)

Ex: ($m = n = 2$)

$$\begin{bmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{bmatrix} + \begin{bmatrix} v_{11} & v_{12} \\ v_{21} & v_{22} \end{bmatrix} = \begin{bmatrix} u_{11} + v_{11} & u_{12} + v_{12} \\ u_{21} + v_{21} & u_{22} + v_{22} \end{bmatrix} \text{ (standard matrix addition)}$$

$$k \begin{bmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{bmatrix} = \begin{bmatrix} ku_{11} & ku_{12} \\ ku_{21} & ku_{22} \end{bmatrix} \text{ (standard scalar multiplication for matrices)}$$

(3) n -th degree or less polynomial space : $V = P_n$

(the set of all real polynomials of degree n or less)

$$p(x) + q(x) = (a_0 + b_0) + (a_1 + b_1)x + \cdots + (a_n + b_n)x^n$$

$$kp(x) = ka_0 + ka_1x + \cdots + ka_nx^n$$

※ By the fact that the set of real numbers is closed under addition and multiplication, it is straightforward to show that P_n satisfies the ten axioms and thus is a vector space

(4) Continuous function space : $V = C(-\infty, \infty)$

(the set of all real-valued continuous functions defined on the entire real line)

$$(f + g)(x) = f(x) + g(x)$$

$$(kf)(x) = kf(x)$$

※ By the fact that the sum of two continuous function is continuous and the product of a scalar and a continuous function is still a continuous function, $C(-\infty, \infty)$ is a vector space

- Summary of important vector spaces

R = set of all real numbers

R^2 = set of all ordered pairs

R^3 = set of all ordered triples

R^n = set of all n -tuples

$C(-\infty, \infty)$ = set of all continuous functions defined on the real number line

$C[a, b]$ = set of all continuous functions defined on a closed interval $[a, b]$

P = set of all polynomials

P_n = set of all polynomials of degree $\leq n$

$M_{m,n}$ = set of $m \times n$ matrices

$M_{n,n}$ = set of $n \times n$ square matrices

※ Each element in a vector space is called a vector, so a vector can be a real number, an n -tuple, a matrix, a polynomial, a continuous function, etc.

- **Theorem 2: Properties of scalar multiplication**

Let $\mathbf{v}$ be any element of a vector space V , and let c be any scalar. Then the following properties are true

(1) $0\mathbf{v} = \mathbf{0}$

(2) $c\mathbf{0} = \mathbf{0}$

(3) If $c\mathbf{v} = \mathbf{0}$, either $c = 0$ or $\mathbf{v} = \mathbf{0}$

(4) $(-1)\mathbf{v} = -\mathbf{v}$ (the additive inverse of $\mathbf{v}$ equals $(-1)\mathbf{v}$)

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- **Notes:** To show that a set is not a vector space, you need only find one axiom that is not satisfied
 - **Ex 1:** The set of all integers is not a vector space

Pf:

$1 \in V$, and $\frac{1}{2}$ is a real-number scalar

$$\begin{array}{c} \left(\frac{1}{2}\right)(1) = \frac{1}{2} \notin V \quad \text{(it is not closed under scalar multiplication)} \\ \uparrow \quad \uparrow \quad \uparrow \\ \text{scalar} \quad \text{integer} \quad \text{noninteger} \end{array}$$

- **Ex 2:** The set of all (exact) second-degree polynomial functions is not a vector space

Pf: Let $p(x) = x^2$ and $q(x) = -x^2 + x + 1$

$$\Rightarrow p(x) + q(x) = x + 1 \notin V$$

(it is not closed under vector addition)

■ Ex 3:

$V = \mathbb{R}^2$ = the set of all ordered pairs of real numbers

vector addition: $(u_1, u_2) + (v_1, v_2) = (u_1 + v_1, u_2 + v_2)$

scalar multiplication: $c(u_1, u_2) = (cu_1, 0)$ (nonstandard definition)

Verify V is not a vector space

Sol:

This kind of setting can satisfy the first nine axioms of the definition of a vector space (you can try to show that), but it violates the tenth axiom

$$\because 1(1, 1) = (1, 0) \neq (1, 1)$$

∴ the set (together with the two given operations) is not a vector space

3 Subspaces of Vector Spaces

- **Subspace :**

$(V, +, \cdot)$: a vector space

$\left. \begin{array}{l} W \neq \Phi \\ W \subseteq V \end{array} \right\}$: a nonempty subset

$(W, +, \cdot)$: The nonempty subset W is called a subspace **if W is a vector space** under the operations of addition and scalar multiplication defined in V

- **Trivial subspace :**

Every vector space V has at least two subspaces

(1) Zero vector space $\{\mathbf{0}\}$ is a subspace of V (It satisfies the ten axioms)

(2) V is a subspace of V

※ Any subspaces other than these two are call proper (or nontrivial) subspaces

- Examination of whether W being a subspace

- Since the operations defined on W are the same as those defined on V , and most of the ten axioms are inherited from the properties for operations, it is not needed to verify these axioms
- Therefore, the following theorem tells us it is sufficient to test for the closure conditions under vector addition and scalar multiplication to identify that a nonempty subset of a vector space is a subspace

- Theorem 3: Test whether a nonempty subset being a subspace

If W is a nonempty subset of a vector space V , then W is a subspace of V if and only if the following conditions hold

(1) If $\mathbf{u}$ and $\mathbf{v}$ are in W , then $\mathbf{u} + \mathbf{v}$ is in W

(2) If $\mathbf{u}$ is in W and c is any scalar, then $c\mathbf{u}$ is in W

■ **Ex 2: A subspace of $M_{2 \times 2}$**

Let W be the set of all 2×2 symmetric matrices. Show that W is a subspace of the vector space $M_{2 \times 2}$, with the standard operations of matrix addition and scalar multiplication

Sol:

First, we know that W , the set of all 2×2 symmetric matrices, is a nonempty subset of the vector space $M_{2 \times 2}$

Second,

$$A_1 \in W, A_2 \in W \Rightarrow (A_1 + A_2)^T = A_1^T + A_2^T = A_1 + A_2 \quad (A_1 + A_2 \in W)$$

$$c \in R, A \in W \Rightarrow (cA)^T = cA^T = cA \quad (cA \in W)$$

The definition of a symmetric matrix A is that $A^T = A$

$\therefore W$ is a subspace of $M_{2 \times 2}$

■ **Ex 3: The set of singular matrices is not a subspace of $M_{2 \times 2}$**

Let W be the set of singular (noninvertible) matrices of order 2. Show that W is not a subspace of $M_{2 \times 2}$ with the standard matrix operations

Sol:

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \in W, \quad B = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \in W$$

$$\therefore A + B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \notin W \quad (W \text{ is not closed under vector addition})$$

$\therefore W_2$ is not a subspace of $M_{2 \times 2}$

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- Ex 4: The set of first-quadrant vectors is not a subspace of R^2

Show that $W = \{(x_1, x_2) : x_1 \geq 0 \text{ and } x_2 \geq 0\}$, with the standard operations, is not a subspace of R^2

Sol:

Let $\mathbf{u} = (1, 1) \in W$

$\therefore (-1)\mathbf{u} = (-1)(1, 1) = (-1, -1) \notin W$
(W is not closed under scalar multiplication)

$\therefore W$ is not a subspace of R^2

▪ **Ex 5: Identify subspaces of R^2**

Which of the following two subsets is a subspace of R^2 ?

(a) The set of points on the line given by $x+2y=0$

(b) The set of points on the line given by $x+2y=1$

Sol:

(a) $W = \{(x, y) \mid x + 2y = 0\} = \{(-2t, t) \mid t \in R\}$ (Note: the zero vector $(0,0)$ is on this line)

Let $\mathbf{v}_1 = (-2t_1, t_1) \in W$ and $\mathbf{v}_2 = (-2t_2, t_2) \in W$

$\therefore \mathbf{v}_1 + \mathbf{v}_2 = (-2(t_1 + t_2), t_1 + t_2) \in W$ (closed under vector addition)

$c\mathbf{v}_1 = (-2(ct_1), ct_1) \in W$ (closed under scalar multiplication)

$\therefore W$ is a subspace of R^2

(b) $W = \{(x, y) \mid x + 2y = 1\}$ (Note: the zero vector $(0, 0)$ is not on this line)

Consider $\mathbf{v} = (1, 0) \in W$

$\because (-1)\mathbf{v} = (-1, 0) \notin W \quad \therefore W$ is not a subspace of R^2

▪ Ex 8: Identify subspaces of R^3

Which of the following subsets is a subspace of R^3 ?

(a) $W = \{(x_1, x_2, 1) \mid x_1, x_2 \in R\}$ (Note: the zero vector is not in W)

(b) $W = \{(x_1, x_1 + x_3, x_3) \mid x_1, x_3 \in R\}$ (Note: the zero vector is in W)

Sol:

(a)

Consider $\mathbf{v} = (0, 0, 1) \in W$

$\because (-1)\mathbf{v} = (0, 0, -1) \notin W$

$\therefore W$ is not a subspace of R^3

(b)

Consider $\mathbf{v} = (v_1, v_1 + v_3, v_3) \in W$ and $\mathbf{u} = (u_1, u_1 + u_3, u_3) \in W$

$$\therefore \mathbf{v} + \mathbf{u} = (v_1 + u_1, (v_1 + u_1) + (v_3 + u_3), v_3 + u_3) \in W$$

$$c\mathbf{v} = (cv_1, (cv_1) + (cv_3), cv_3) \in W$$

$\therefore W$ is closed under addition and scalar multiplication,
so W is a subspace of R^3

- **Theorem 4: The intersection of two subspaces is a subspace**

If V and W are both subspaces of a vector space U ,
then the intersection of V and W (denoted by $V \cap W$)
is also a subspace of U

Pf:

- (1) For $\mathbf{v}_1$ and $\mathbf{v}_2$ in $V \cap W$, since $\mathbf{v}_1$ and $\mathbf{v}_2$ are in V (and W),
 $\mathbf{v}_1 + \mathbf{v}_2$ is in V (and W). Therefore, $\mathbf{v}_1 + \mathbf{v}_2$ is in $V \cap W$
- (2) For $\mathbf{v}_1$ in $V \cap W$, since $\mathbf{v}_1$ is in V (and W),
 $c\mathbf{v}_1$ is in V (and W). Therefore, $c\mathbf{v}_1$ is in $V \cap W$

Consequently, we can conclude the intersection of V and W
($V \cap W$) is also a subspace of U

※ However, the union of two subspaces is not a subspace

4.4 Spanning Sets and Linear Independence

- **Linear combination :**

A vector $\mathbf{u}$ in a vector space V is called a linear combination of the vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$ in V if $\mathbf{u}$ can be written in the form

$$\mathbf{u} = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \dots + c_k \mathbf{v}_k,$$

where $c_1, c_2, \dots, c_k$ are real-number scalars

▪ **Ex : Finding a linear combination**

$$\mathbf{v}_1 = (1, 2, 3) \quad \mathbf{v}_2 = (0, 1, 2) \quad \mathbf{v}_3 = (-1, 0, 1)$$

Prove (a) $\mathbf{w} = (1, 1, 1)$ is a linear combination of $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$

(b) $\mathbf{w} = (1, -2, 2)$ is not a linear combination of $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$

Sol:

$$(a) \mathbf{w} = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + c_3 \mathbf{v}_3$$

$$\begin{aligned} (1, 1, 1) &= c_1(1, 2, 3) + c_2(0, 1, 2) + c_3(-1, 0, 1) \\ &= (c_1 - c_3, 2c_1 + c_2, 3c_1 + 2c_2 + c_3) \end{aligned}$$

$$\begin{aligned} c_1 - c_3 &= 1 \\ \Rightarrow 2c_1 + c_2 &= 1 \\ 3c_1 + 2c_2 + c_3 &= 1 \end{aligned}$$

$$\Rightarrow \left[\begin{array}{ccc|c} 1 & 0 & -1 & 1 \\ 2 & 1 & 0 & 1 \\ 3 & 2 & 1 & 1 \end{array} \right] \xrightarrow{\text{G.-J. E.}} \left[\begin{array}{ccc|c} 1 & 0 & -1 & 1 \\ 0 & 1 & 2 & -1 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$\Rightarrow c_1 = 1 + t, \quad c_2 = -1 - 2t, \quad c_3 = t$$

(this system has infinitely many solutions)

$$\stackrel{t=1}{\Rightarrow} \mathbf{w} = 2\mathbf{v}_1 - 3\mathbf{v}_2 + \mathbf{v}_3$$

$$\stackrel{t=2}{\Rightarrow} \mathbf{w} = 3\mathbf{v}_1 - 5\mathbf{v}_2 + 2\mathbf{v}_3$$
$$\vdots$$

(b)

$$\mathbf{w} = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + c_3 \mathbf{v}_3$$

$$\Rightarrow \left[\begin{array}{ccc|c} 1 & 0 & -1 & 1 \\ 2 & 1 & 0 & -2 \\ 3 & 2 & 1 & 2 \end{array} \right] \xrightarrow{\text{G.-J. E.}} \left[\begin{array}{ccc|c} 1 & 0 & -1 & 1 \\ 0 & 1 & 2 & -4 \\ 0 & 0 & 0 & 7 \end{array} \right]$$

$\Rightarrow$ This system has no solution since the third row means

$$0 \cdot c_1 + 0 \cdot c_2 + 0 \cdot c_3 = 7$$

$\Rightarrow \mathbf{w}$ can not be expressed as $c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + c_3 \mathbf{v}_3$

- **The span of a set: $\text{span}(S)$**

If $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ is a set of vectors in a vector space V , then the span of S is the set of all linear combinations of the vectors in S ,

$$\text{span}(S) = \{c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \dots + c_k\mathbf{v}_k \mid \forall c_i \in R\}$$

(the set of all linear combinations of vectors in S)

- **Definition of a spanning set of a vector space:**

If every vector in a given vector space V can be written as a linear combination of vectors in a set S , then S is called a **spanning set** of the vector space V

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- **Note:** The above statement can be expressed as follows

$$\text{span}(S) = V$$

$$\Leftrightarrow S \text{ spans (generates) } V$$

$$\Leftrightarrow V \text{ is spanned (generated) by } S$$

$$\Leftrightarrow S \text{ is a spanning set of } V$$

- **Ex 4:**

(a) The set $S = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ spans R^3 because any vector

$\mathbf{u} = (u_1, u_2, u_3)$ in R^3 can be written as

$$\mathbf{u} = u_1(1, 0, 0) + u_2(0, 1, 0) + u_3(0, 0, 1)$$

(b) The set $S = \{1, x, x^2\}$ spans P_2 because any polynomial function

$p(x) = a + bx + cx^2$ in P_2 can be written as

$$p(x) = a(1) + b(x) + c(x^2)$$

■ Ex 5: A spanning set for R^3

Show that the set $S = \{(1, 2, 3), (0, 1, 2), (-2, 0, 1)\}$ spans R^3

Sol:

We must determine whether an arbitrary vector $\mathbf{u} = (u_1, u_2, u_3)$ in R^3 can be expressed as a linear combination of $\mathbf{v}_1 = (1, 2, 3)$, $\mathbf{v}_2 = (0, 1, 2)$, and $\mathbf{v}_3 = (-2, 0, 1)$

$$\begin{aligned} \text{If } \mathbf{u} = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + c_3 \mathbf{v}_3 &\Rightarrow \begin{aligned} c_1 - 2c_3 &= u_1 \\ 2c_1 + c_2 &= u_2 \\ 3c_1 + 2c_2 + c_3 &= u_3 \end{aligned} \end{aligned}$$

The above problem thus reduces to determining whether this system is consistent for all values of u_1 , u_2 , and u_3

$$\because |A| = \begin{vmatrix} 1 & 0 & -2 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{vmatrix} \neq 0$$

$\therefore A\mathbf{x} = \mathbf{u}$ has exactly one solution for every $\mathbf{u}$

$$\Rightarrow \text{span}(S) = R^3$$

▪ **Theorem 5: $\text{span}(S)$ is a subspace of V**

If $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ is a set of vectors in a vector space V , then

- (a) $\text{span}(S)$ is a subspace of V
- (b) $\text{span}(S)$ is the smallest subspace of V that contains S , i.e., every other subspace of V containing S must contain $\text{span}(S)$

- **Definitions of Linear Independence (L.I.) and Linear Dependence (L.D.) :**

$S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$: a set of vectors in a vector space V

For $c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \dots + c_k\mathbf{v}_k = \mathbf{0}$

(1) If the equation has only the trivial solution ($c_1 = c_2 = \dots = c_k = 0$)

then S (or $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$) is called **linearly independent**

(2) If the equation has a nontrivial solution (i.e., not all zeros),

then S (or $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$) is called **linearly dependent** (The name of linear dependence is from the fact that in this case, there exist a $\mathbf{v}_i$ which can be represented by the linear combination of $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_{i-1}, \mathbf{v}_{i+1}, \dots, \mathbf{v}_k\}$ in which the coefficients are not all zero.

■ Ex 8: Testing for linear independence

Determine whether the following set of vectors in R^3 is L.I. or L.D.

$$S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\} = \{(1, 2, 3), (0, 1, 2), (-2, 0, 1)\}$$

Sol:

$$\begin{aligned} c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + c_3 \mathbf{v}_3 = \mathbf{0} &\Rightarrow \begin{aligned} c_1 - 2c_3 &= 0 \\ 2c_1 + c_2 &= 0 \\ 3c_1 + 2c_2 + c_3 &= 0 \end{aligned} \end{aligned}$$

$$\Rightarrow \left[\begin{array}{ccc|c} 1 & 0 & -2 & 0 \\ 2 & 1 & 0 & 0 \\ 3 & 2 & 1 & 0 \end{array} \right] \xrightarrow{\text{G.-J. E.}} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array} \right]$$

$$\Rightarrow c_1 = c_2 = c_3 = 0 \quad (\text{only the trivial solution})$$

(or $\det(A) = -1 \neq 0$, so there is only the trivial solution)

$\Rightarrow S$ is (or $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ are) linearly independent

- Ex 9: Testing for linear independence

Determine whether the following set of vectors in P_2 is L.I. or L.D.

$$S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\} = \{1+x-2x^2, 2+5x-x^2, x+x^2\}$$

Sol:

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + c_3\mathbf{v}_3 = \mathbf{0}$$

$$\text{i.e., } c_1(1+x-2x^2) + c_2(2+5x-x^2) + c_3(x+x^2) = 0+0x+0x^2$$

$$\begin{array}{rcl} c_1 + 2c_2 & = & 0 \\ c_1 + 5c_2 + c_3 & = & 0 \\ -2c_1 - c_2 + c_3 & = & 0 \end{array} \quad \left[\begin{array}{ccc|c} 1 & 2 & 0 & 0 \\ 1 & 5 & 1 & 0 \\ -2 & -1 & 1 & 0 \end{array} \right] \xrightarrow{\text{G. E.}} \left[\begin{array}{ccc|c} 1 & 2 & 0 & 0 \\ 0 & 1 & 1/3 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

This system has infinitely many solutions

(i.e., this system has nontrivial solutions, e.g., $c_1=2$, $c_2=-1$, $c_3=3$)

S is (or $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ are) linearly dependent

- Ex 10: Testing for linear independence

Determine whether the following set of vectors in the 2×2 matrix space is L.I. or L.D.

$$S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\} = \left\{ \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 3 & 0 \\ 2 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 2 & 0 \end{bmatrix} \right\}$$

Sol:

$$c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + c_3 \mathbf{v}_3 = \mathbf{0}$$

$$c_1 \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix} + c_2 \begin{bmatrix} 3 & 0 \\ 2 & 1 \end{bmatrix} + c_3 \begin{bmatrix} 1 & 0 \\ 2 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$2c_1 + 3c_2 + c_3 = 0$$

$$c_1 = 0$$

$$2c_2 + 2c_3 = 0$$

$$c_1 + c_2 = 0$$

$$\left[\begin{array}{ccc|c} 2 & 3 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 2 & 2 & 0 \\ 1 & 1 & 0 & 0 \end{array} \right] \xrightarrow{\text{G.-J. E.}} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$c_1 = c_2 = c_3 = 0$ (This system has only the trivial solution)

S is linearly independent

- Theorem 6: A property of linearly dependent sets

A set $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ is linearly dependent if and only if at least one of the vectors $\mathbf{v}_i$ in S can be written as a linear combination of the other vectors in S

Pf:

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \dots + c_k\mathbf{v}_k = \mathbf{0}$$

$\because S$ is linearly dependent (there exist nontrivial solutions)

For $c_i \neq 0$ for some i

$$\Rightarrow \mathbf{v}_i = \left(-\frac{c_1}{c_i}\right)\mathbf{v}_1 + \dots + \left(-\frac{c_{i-1}}{c_i}\right)\mathbf{v}_{i-1} + \left(-\frac{c_{i+1}}{c_i}\right)\mathbf{v}_{i+1} + \dots + \left(-\frac{c_k}{c_i}\right)\mathbf{v}_k$$

4.41

$(\Leftarrow)$

Let $\mathbf{v}_i = d_1\mathbf{v}_1 + \dots + d_{i-1}\mathbf{v}_{i-1} + d_{i+1}\mathbf{v}_{i+1} + \dots + d_k\mathbf{v}_k$

$$d_1\mathbf{v}_1 + \dots + d_{i-1}\mathbf{v}_{i-1} - \mathbf{v}_i + d_{i+1}\mathbf{v}_{i+1} + \dots + d_k\mathbf{v}_k = \mathbf{0}$$

$$c_1 = d_1, c_2 = d_2, \dots, c_i = -1, \dots, c_k = d_k \quad \text{(there exists at least this nontrivial solution)}$$

$\Rightarrow S$ is linearly dependent

- **Corollary to Theorem 4.8:**

Two vectors $\mathbf{u}$ and $\mathbf{v}$ in a vector space V are linearly dependent (for $k = 2$ in Theorem 6) if and only if one is a scalar multiple of the other

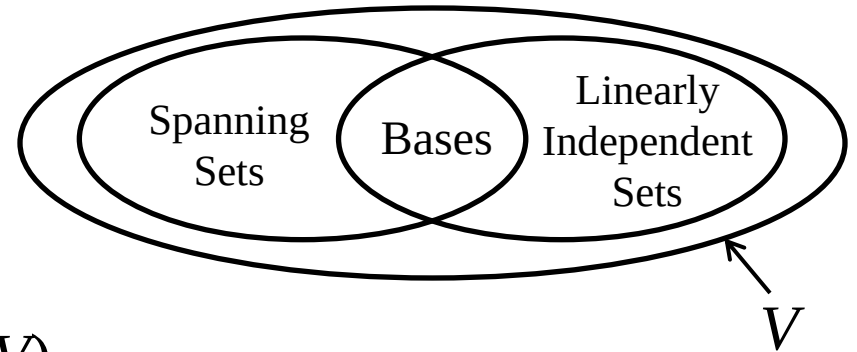
4.5 Basis and Dimension

- **Basis :**

V : a vector space

$S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ in V

- (1) S spans V (i.e., $\text{span}(S) = V$)
(For any $\mathbf{u} \in V$, $\sum c_i \mathbf{v}_i = A\mathbf{x} = \mathbf{u}$ has a solution ($\det(A) \neq 0$),
see Ex 5 on Slide 4.44)
- (2) S is linearly independent
(For $\sum c_i \mathbf{v}_i = A\mathbf{x} = \mathbf{0}$, there is only the trivial solution ($\det(A) \neq 0$),



Then, S is called a basis for V

- **Notes:**

A basis S must have enough vectors to span V , but not so many vectors that one of them could be written as a linear combination of the other vectors in S

■ **Notes:**

(1) the **standard basis** for R^3 :

$$\{i, j, k\} \quad i = (1, 0, 0), j = (0, 1, 0), k = (0, 0, 1)$$

(2) the **standard basis** for R^n :

$$\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\} \quad \mathbf{e}_1=(1,0,\dots,0), \mathbf{e}_2=(0,1,\dots,0),\dots, \mathbf{e}_n=(0,0,\dots,1)$$

Ex: For R^4 , $\{(1,0,0,0), (0,1,0,0), (0,0,1,0), (0,0,0,1)\}$

(3) the **standard basis** matrix space:

Ex: 2×2 matrix space:

$$\left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$$

(4) the **standard basis** for $P_n(x)$:

$$\{1, x, x^2, \dots, x^n\}$$

Ex: $P_3(x) \quad \{1, x, x^2, x^3\}$

■ **Ex 2: The nonstandard basis for R^2**

Show that $S = \{\mathbf{v}_1, \mathbf{v}_2\} = \{(1, 1), (1, -1)\}$ is a basis for R^2

$$(1) \text{ For any } \mathbf{u} = (u_1, u_2) \in R^2, c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 = \mathbf{u} \Rightarrow \begin{cases} c_1 + c_2 = u_1 \\ c_1 - c_2 = u_2 \end{cases}$$

Because the coefficient matrix of this system has a **nonzero determinant**, the system has a unique solution for each $\mathbf{u}$. Thus you can conclude that S spans R^2

$$(2) \text{ For } c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 = \mathbf{0} \Rightarrow \begin{cases} c_1 + c_2 = 0 \\ c_1 - c_2 = 0 \end{cases}$$

Because the coefficient matrix of this system has a **nonzero determinant**, you know that the system has only the trivial solution. Thus you can conclude that S is linearly independent

According to the above two arguments, we can conclude that S is a (nonstandard) basis for R^2

- **Theorem 7: Uniqueness of basis representation for any vectors**

If $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ is a basis for a vector space V , then every vector in V can be written in one and only one way as a linear combination of vectors in S

Pf:

$$\because S \text{ is a basis} \Rightarrow \begin{cases} (1) \text{ span}(S) = V \\ (2) S \text{ is linearly independent} \end{cases}$$

$$\because \text{span}(S) = V \quad \text{Let } \mathbf{v} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \dots + c_n\mathbf{v}_n$$

$$\mathbf{v} = b_1\mathbf{v}_1 + b_2\mathbf{v}_2 + \dots + b_n\mathbf{v}_n$$

$$\square \mathbf{v} + (-1)\mathbf{v} = \mathbf{0} = (c_1 - b_1)\mathbf{v}_1 + (c_2 - b_2)\mathbf{v}_2 + \dots + (c_n - b_n)\mathbf{v}_n$$

$\because S$ is linearly independent $\Rightarrow$ with only trivial solution
 $\Rightarrow$ coefficients for $\mathbf{v}_i$ are all zero

$$\square c_1 = b_1, c_2 = b_2, \dots, c_n = b_n \quad (\text{i.e., unique basis representation})_{4.47}$$

■ Theorem 8: Bases and linear dependence

If $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ is a basis for a vector space V , then every set containing more than n vectors in V is linearly dependent (In other words, every linearly independent set contains at most n vectors)

Pf:

Let $S_1 = \{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_m\}$, $m > n$

$\because \text{span}(S) = V$

$$\begin{array}{lcl} & & \mathbf{u}_1 = c_{11}\mathbf{v}_1 + c_{21}\mathbf{v}_2 + \dots + c_{n1}\mathbf{v}_n \\ \mathbf{u}_i \in V & \Rightarrow & \mathbf{u}_2 = c_{12}\mathbf{v}_1 + c_{22}\mathbf{v}_2 + \dots + c_{n2}\mathbf{v}_n \\ & & \vdots \\ & & \mathbf{u}_m = c_{1m}\mathbf{v}_1 + c_{2m}\mathbf{v}_2 + \dots + c_{nm}\mathbf{v}_n \end{array}$$

- Theorem 9: Number of vectors in a basis

If a vector space V has one basis with n vectors, then every basis for V has n vectors

Pf: ※ According to Thm.8, every linearly independent set contains at most n vectors in a vector space if there is a set of n vectors spanning that vector space

$S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$
 $S' = \{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_m\}$ are two bases with different sizes for V

$$\left. \begin{array}{l} S \text{ is a basis spanning } V \\ S' \text{ is a set of L.I. vectors} \end{array} \right\} \Rightarrow m \leq n$$
$$\left. \begin{array}{l} S' \text{ is a basis spanning } V \\ S \text{ is a set of L.I. vectors} \end{array} \right\} \Rightarrow n \leq m$$
$$\left. \begin{array}{l} \Rightarrow m \leq n \\ \Rightarrow n \leq m \end{array} \right\} \Rightarrow n = m$$

※ For R^3 , since the standard basis $\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ can span this vector space, you can infer any basis that can span R^3 must have exactly 3 vectors

※ For example, $S = \{(1, 2, 3), (0, 1, 2), (-2, 0, 1)\}$ is another basis of R^3 (because S can span R^3 and S is linearly independent), and S has 3 vectors

- **Dimension:**

The dimension of a vector space V is defined to be the number of vectors in a basis for V

V : a vector space S : a basis for V

$$\dim(V) = \#(S) \quad (\text{the number of vectors in a basis } S)$$

- **Finite dimensional:**

A vector space V is finite dimensional if it has a basis consisting of a finite number of elements

- **Infinite dimensional:**

If a vector space V is not finite dimensional, then it is called infinite dimensional

■ Ex 9: Finding the dimension of a subspace of R^3

(a) $W = \{(d, c-d, c) : c \text{ and } d \text{ are real numbers}\}$

(b) $W = \{(2b, b, 0) : b \text{ is a real number}\}$

Sol: (Hint: find a set of L.I. vectors that spans the subspace, i.e., find a basis for the subspace.)

(a) $(d, c-d, c) = c(0, 1, 1) + d(1, -1, 0)$

$$S = \{(0, 1, 1), (1, -1, 0)\} \quad (S \text{ is L.I. and } S \text{ spans } W)$$

S is a basis for W

$$\dim(W) = \#(S) = 2$$

(b) $\because (2b, b, 0) = b(2, 1, 0)$

$$S = \{(2, 1, 0)\} \text{ spans } W \text{ and } S \text{ is L.I.}$$

S is a basis for W

$$\dim(W) = \#(S) = 1$$

▪ **Ex 11: Finding the dimension of a subspace of $M_{2 \times 2}$**

Let W be the subspace of all symmetric matrices in $M_{2 \times 2}$.

What is the dimension of W ?

Sol:

$$W = \left\{ \begin{bmatrix} a & b \\ b & c \end{bmatrix} \mid a, b, c \in R \right\}$$

$$\because \begin{bmatrix} a & b \\ b & c \end{bmatrix} = a \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + b \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} + c \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\Rightarrow S = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\} \text{ spans } W \text{ and } S \text{ is L.I.}$$

$$S \text{ is a basis for } W \quad \dim(W) = \#(S) = 3$$

ROW SPACE AND COLOUMN SPACE

$m \times n$ matrix A can be abbreviated by the row vectors or column vectors as follows:

$$\begin{aligned} A &= \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} = \begin{bmatrix} \mathbf{r}_1 \\ \mathbf{r}_2 \\ \vdots \\ \mathbf{r}_m \end{bmatrix} \\ &= \begin{bmatrix} \mathbf{c}_1 & \mathbf{c}_2 & \cdots & \mathbf{c}_n \end{bmatrix}, \end{aligned}$$

where the $\mathbf{r}_i$'s are the row vectors of A that are in $\mathbb{R}^n$, and the $\mathbf{c}_j$'s are the column vectors of A that are in $\mathbb{R}^m$.

DEFINITIONS

Definition 3.7 Let A be an $m \times n$ matrix with row vectors $\{\mathbf{r}_1, \dots, \mathbf{r}_m\}$ and column vectors $\{\mathbf{c}_1, \dots, \mathbf{c}_n\}$.

- (1) The **row space** of A is the subspace in $\mathbb{R}^n$ spanned by the row vectors $\{\mathbf{r}_1, \dots, \mathbf{r}_m\}$, denoted by $\mathcal{R}(A)$.
- (2) The **column space** of A is the subspace in $\mathbb{R}^m$ spanned by the column vectors $\{\mathbf{c}_1, \dots, \mathbf{c}_n\}$, denoted by $\mathcal{C}(A)$.
- (3) The solution set of the homogeneous equation $A\mathbf{x} = \mathbf{0}$ is called the **null space** of A , denoted by $\mathcal{N}(A)$.

Note that the null space $\mathcal{N}(A)$ is a subspace of the n -space $\mathbb{R}^n$, whose dimension is called the **nullity** of A . Since the row vectors of A are just the column vectors of its transpose A^T , and the column vectors of A are the row vectors of A^T , the row space of A is the column space of A^T ; that is,

$$\mathcal{R}(A) = \mathcal{C}(A^T) \quad \text{and} \quad \mathcal{C}(A) = \mathcal{R}(A^T).$$

EXAMPLES

Example 3.19 Let U be in a reduced row-echelon form given as

$$U = \begin{bmatrix} 1 & 0 & 0 & 2 & 2 \\ 0 & 1 & 0 & -1 & 3 \\ 0 & 0 & 1 & 4 & -1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Clearly, the first three nonzero row vectors containing leading 1's are linearly independent and they form a basis for the row space $\mathcal{R}(U)$, so that $\dim \mathcal{R}(U) = 3$. On the other hand, note that the first three columns containing leading 1's are linearly independent (see Theorem 3.5), and that the last two column vectors can be expressed as linear combinations of them. Hence, they form a basis for $\mathcal{C}(U)$, and $\dim \mathcal{C}(U) = 3$. To find a basis for the null space $\mathcal{N}(U)$, we first solve the system $U\mathbf{x} = \mathbf{0}$ with arbitrary values s and t for the free variables x_4 and x_5 , and get the solution

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} -2s & - & 2t \\ s & - & 3t \\ -4s & + & t \\ s & & \\ & & t \end{bmatrix} = s \begin{bmatrix} -2 \\ 1 \\ -4 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} -2 \\ -3 \\ 1 \\ 0 \\ 1 \end{bmatrix} = s\mathbf{n}_s + t\mathbf{n}_t,$$

where $\mathbf{n}_s = (-2, 1, -4, 1, 0)$, $\mathbf{n}_t = (-2, -3, 1, 0, 1)$. It shows that these two vectors $\mathbf{n}_s$ and $\mathbf{n}_t$ span the null space $\mathcal{N}(U)$, and they are clearly linearly independent. Hence, the set $\{\mathbf{n}_s, \mathbf{n}_t\}$ is a basis for the null space $\mathcal{N}(U)$. $\square$

Example 3.20 Let A be a matrix given as

$$A = \begin{bmatrix} 1 & 2 & 0 & 2 & 5 \\ -2 & -5 & 1 & -1 & -8 \\ 0 & -3 & 3 & 4 & 1 \\ 3 & 6 & 0 & -7 & 2 \end{bmatrix} = \begin{bmatrix} \mathbf{r}_1 \\ \mathbf{r}_2 \\ \mathbf{r}_3 \\ \mathbf{r}_4 \end{bmatrix}.$$

Find bases for the row space $\mathcal{R}(A)$, the null space $\mathcal{N}(A)$, and the column space $\mathcal{C}(A)$ of A .

Solution: (1) Find a basis for $\mathcal{R}(A)$: By Gauss-Jordan elimination on A , we get the reduced row-echelon form U :

$$U = \begin{bmatrix} 1 & 0 & 2 & 0 & 1 \\ 0 & 1 & -1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Since the three nonzero row vectors

$$\begin{aligned}\mathbf{v}_1 &= (1, 0, 2, 0, 1), \\ \mathbf{v}_2 &= (0, 1, -1, 0, 1), \\ \mathbf{v}_3 &= (0, 0, 0, 1, 1)\end{aligned}$$

of U are linearly independent, they form a basis for the row space $\mathcal{R}(U) = \mathcal{R}(A)$, so $\dim \mathcal{R}(A) = 3$. (Note that in the process of Gaussian elimination, we did not use a permutation matrix. This means that the three nonzero rows of U were obtained from the first three row vectors $\mathbf{r}_1, \mathbf{r}_2, \mathbf{r}_3$ of A and the fourth row $\mathbf{r}_4$ of A turned out to be a linear combination of them. Thus the first three row vectors of A also form a basis for the row space.)

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} -2s & -t \\ s & -t \\ s & \\ & -t \\ & t \end{bmatrix} = s \begin{bmatrix} -2 \\ 1 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ -1 \\ 0 \\ -1 \\ 1 \end{bmatrix} = s\mathbf{n}_s + t\mathbf{n}_t,$$

where $\mathbf{n}_s = (-2, 1, 1, 0, 0)$ and $\mathbf{n}_t = (-1, -1, 0, -1, 1)$. In fact, the two vectors $\mathbf{n}_s$ and $\mathbf{n}_t$ are the solutions when the values of $(x_3, x_5) = (s, t)$ are $(1, 0)$ and those of $(x_3, x_5) = (s, t)$ are $(0, 1)$, respectively. They must be linearly independent, since $(1, 0)$ and $(0, 1)$, as the (x_3, x_5) -coordinates of $\mathbf{n}_s$ and $\mathbf{n}_t$ respectively, are clearly linearly independent. Since any solution of $U\mathbf{x} = \mathbf{0}$ is a linear combination of them, the set $\{\mathbf{n}_s, \mathbf{n}_t\}$ is a basis for the null space $\mathcal{N}(U) = \mathcal{N}(A)$. Thus $\dim \mathcal{N}(A) = 2 = \text{the number of free variables in } U\mathbf{x} = \mathbf{0}$.

In summary, given a matrix A , we first find the (reduced) row-echelon form U of A by Gauss-Jordan elimination. Then a basis for $\mathcal{R}(A) = \mathcal{R}(U)$ is the set of nonzero rows vectors of U , and a basis for $\mathcal{N}(A) = \mathcal{N}(U)$ can be found by solving $U\mathbf{x} = \mathbf{0}$, which is easy. On the other hand, one has to be careful for $\mathcal{C}(U) \neq \mathcal{C}(A)$ in general, since the column space of A is not preserved by Gauss-Jordan elimination. However, we have $\dim \mathcal{C}(A) = \dim \mathcal{C}(U)$, and a basis for $\mathcal{C}(A)$ can be selected from the column vectors in A , not in U , as those corresponding to the basic variables (or the leading 1's in U). To show that those column vectors indeed form a basis for $\mathcal{C}(A)$, we used a basis for the null space $\mathcal{N}(A)$ to eliminate the redundant columns.

Note that a basis for the column space $\mathcal{C}(A)$ can be also found with the elementary column operations, which is the same as finding a basis for the row space $\mathcal{R}(A^T)$ of A^T .

Theorem 3.14 *Let U be a (reduced) row-echelon form of a matrix A . Then*

$$\mathcal{R}(A) = \mathcal{R}(U) \quad \text{and} \quad \mathcal{N}(A) = \mathcal{N}(U).$$

Moreover, if U has r nonzero row vectors containing leading 1's, then they form a basis for the row space $\mathcal{R}(A)$, so that the dimension of $\mathcal{R}(A)$ is r .

Theorem 3.15 (The first fundamental theorem) *For any $m \times n$ matrix A , the row space and the column space of A have the same dimension; that is, $\dim \mathcal{R}(A) = \dim \mathcal{C}(A)$.*

Corollary 3.17 *For any $m \times n$ matrix A ,*

$$\begin{aligned}\dim \mathcal{R}(A) + \dim \mathcal{N}(A) &= \text{rank } A + \text{nullity of } A = n, \\ \dim \mathcal{C}(A) + \dim \mathcal{N}(A^T) &= \text{rank } A + \text{nullity of } A^T = m.\end{aligned}$$

Example 3.21 For a 4×5 matrix

$$A = \begin{bmatrix} 1 & 2 & 0 & 2 & 1 \\ -1 & -2 & 1 & 1 & 0 \\ 1 & 2 & -3 & -7 & 2 \\ 1 & 2 & -2 & -4 & 3 \end{bmatrix},$$

by Gaussian elimination, we get

$$U = \begin{bmatrix} 1 & 2 & 0 & 2 & 1 \\ 0 & 0 & 1 & 3 & 1 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

The first three nonzero rows containing leading 1's in U form a basis for $\mathcal{R}(U) = \mathcal{R}(A)$. Note that x_1 , x_3 and x_5 are the basic variables in $U\mathbf{x} = \mathbf{0}$, since the first, third and fifth columns of U contain leading 1's. Thus the three columns $\mathbf{c}_1 = (1, -1, 1, 1)$, $\mathbf{c}_3 = (0, 1, -3, -2)$ and $\mathbf{c}_5 = (1, 0, 2, 3)$ of A , not the three columns in U , corresponding to those basic variables x_1 , x_3 and x_5 form a basis for $\mathcal{C}(A)$. Therefore, $\text{rank } A = \dim \mathcal{R}(A) = \dim \mathcal{C}(A) = 3$, the nullity of $A = \dim \mathcal{N}(A) = 2$, and $\dim \mathcal{N}(A^T) = 1$. $\square$